

Block 2

Money

Unit 4	Role of Money in Modern Economy
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Unit 6	Money Supply

UNIT 4 THE ROLE OF MONEY IN A MODERN ECONOMY

Structure

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4.0 OBJECTIVES

After going through the unit, you will be able to:

- Define money;
- Describe the nature and functions of money;
- Explain measures of money supply;
- Discuss the payment system in a modern economy; and
- Analyse the role of money and credit in the macroeconomy.

4.1 INTRODUCTION

In block-1, you have studied the nature of financial institutions and markets, economic agents in the financial system and also about financial instruments. In this block, we discuss the basic medium of exchange in the economy, namely, money. We will see that money may not be a proper 'financial instrument' but is required for transactions. The present unit, which is the first in the block-2, discusses the basic nature of money. The unit tries to explain what money is, and what is meant by 'near-monies', financial instruments which seem to be quite like money, but do not perform all the functions of money. The unit discusses the functions that money performs and its role in the modern economy. What is the basic structure of payment settlements in a modern economy? How are payments made in transactions across the entire economy? The unit also discusses the basic payment system of an economy that is monetarily and financially mature. Finally, you will be

presented with a discussion regarding the role and impact of money, debt and credit in the macroeconomy. The structure of the unit is as follows. Section 4.2 discusses the nature and functions of money. We discuss various measures of Money Supply in Section 4.3. and Section 4.4 describes the payment system of an economy, with the payment system of India used as an illustration. Finally, Section 4.5 discusses the relation between money and credit, and the role they have in the financial system and the macroeconomy.

4.2 NATURE AND FUNCTIONS OF MONEY

Money is anything that is generally acceptable as a means of payment in the settlement of transactions, including debt.

Money can be referred to as one of the greatest innovations of human society. Money is not needed for its own sake as one needs food, clothes and a living house. Money is needed to mediate transactions. It has purchasing power which enables us to exchange goods and services. So, this makes money a unique commodity.

By reducing transaction costs and thereby encouraging more trade, it can be argued that money allows the development of specialization of labour that is necessary for rapid economic growth. Thus, without money, all must be largely self-sufficient as happens in primitive and tribal societies or in barter economy.

Today, we are all familiar with money in our day-to-day transactions. Perhaps the oldest and simplest role of money has been the 'medium of exchange' for all economic transactions as it is acceptable to everybody. Besides this money is the commonly used means of transferring purchasing power. It does not need to be converted into something else before it can be spent or used for the settlement of debt. What makes money 'money' is the belief held by everyone that it will be accepted as such by all others in the economy.

In all money serves **four** important functions for any society: **Medium of Exchange, Unit of Value, Standard of Deferred Payment, and Store of Value.**

Defining Money: Anything that functions as a means of payment (medium of exchange), unit of account, and store of value. Today money also acts as a legal tender which enhances the trust of people in it.

Measuring Money is not as easy as defining it. What, in your opinion, should we use as money?

- Just currency?
- Currency and other stored forms of money?

We find there are multiple definitions of money and what we call money has changed over time as technology and financial instruments have evolved.

Near Money Definition

Highly liquid assets which are not cash but can easily be converted into cash are referred to as near money assets.

Various sources provide the following examples of near money:

- Money market funds
- Bank and Post Office deposits
- Government treasury securities
- Bonds near their redemption date
- Foreign currencies, especially widely traded ones such as the Euro, Dollar, Pound Sterling or Yen
- Equities/shares of the corporate sector

Money has four primary functions in any economy: as a medium of exchange, as a unit of account, as a store of value and standard of deferred value. Of the four functions, its function as a medium of exchange is what distinguishes money from other assets such as stocks, bonds, and houses.

Medium of Exchange Function

Money as a medium of exchange may be used for any transactions wherein goods or services are purchased or sold. Money as a unit of account can be used to value goods or services and express it in monetary terms. Money can also be stored or conserved for future purposes.

In almost all market transactions in our economy, money in the form of currency or cheques is a medium of exchange; it is used to pay for goods and services. The use of money as a medium of exchange promotes economic efficiency by minimizing the time spent in exchanging goods and services.

Let's consider a *barter economy*, one that is without money, in which goods and services are exchanged directly for other goods and services. The time spent trying to exchange goods or services is called a *transaction cost*. In a barter economy, transaction costs are high because people have to satisfy a "double coincidence of wants"—they have to find someone who has a good or service they want and who also wants the good or service they have to offer.

Money promotes economic efficiency by eliminating much of the time spent exchanging goods and services. It also promotes efficiency by allowing people to specialize in what they do best. Money is, therefore, essential in an economy: it allows the economy to run more smoothly by lowering transaction costs, thereby encouraging specialization and division of labour.

The need for money is so strong that almost every society creates it. For a commodity to function effectively as money, it has to meet several criteria: (1) It must be easily standardized, making it simple to ascertain its value; (2) it must be widely accepted; (3) it must be divisible so that it is easy to “make change”; (4) it must be easy to carry; and (5) it must not deteriorate quickly. Objects that have satisfied these criteria have taken many unusual forms throughout human history.

Unit of Account Function

The second role of money is to provide a unit of account; that is, money is used to measure value in an economy. We measure the value of goods and services in terms of money, just as we measure weight in terms of pounds or distance in terms of miles. To see why this function is important, let's look again at a barter economy, in which money does not perform this function. If the economy has only three goods X, Y, Z then we need to know only three prices to tell us how to exchange one for another: the price of X in terms of Y (that is, how many Ys one has to pay for an X), the price of Y in terms of Z and the price of Z in terms of X. As the number of goods increases in this economy the number of relative prices goes up tremendously.

The solution to the problem is to introduce money into the economy and have all prices quoted in terms of units of that money. We can see that using money as a unit of account lowers transaction costs in an economy by reducing the number of prices that need to be considered. The benefits of this function of money grow as the economy becomes more complex.

Store of Value Function

The store-of-value function of money pertains to its capacity to retain purchasing power over time. In other words, money serves as a means to hold wealth in a stable form that can be readily accessed and utilized in the future. People use money to preserve the economic value of their assets and savings.

However, the effectiveness of money as a store of value can fluctuate due to factors such as inflation, economic stability, and the availability of alternative investment options. Inflation, for example, can diminish the value of money over time, reducing its purchasing power. In such scenarios, individuals might seek alternative stores of value, like real estate, precious metals, stocks, or bonds, which may offer better protection against inflation or higher returns.

Despite these challenges, money remains a widely accepted and convenient store of value because of its liquidity, divisibility, and broad acceptance of goods and services. Although it may not always preserve wealth as effectively as other assets, its ease of use and stability make it a crucial element of modern economies.

Check Your Progress 1

- 1) Discuss some limitations of the Barter system.

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- 2) What is Near money? Give some examples of Near money.

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- 3) Briefly explain the various functions of Money.

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4.3 MEASURES OF MONEY SUPPLY

There is a vast array of financial assets in any economy, from currency to complicated claims on other financial assets. Which of these assets should be called money? There are two types of measures of money: narrow money and broad money. Narrow money comprises those claims that can be used directly, instantly, and without restrictions to make payments. These claims are *liquid*. Broad money, includes, in addition, claims that are not instantly liquid — withdrawal of time deposits, for example, may require notice to the depository institution; money market mutual funds may set a minimum on the size of cheques drawn on an account.

The money supply is the total amount of money available in an economy at a particular point in time (a stock).

Usually, the money stock in an economy is measured as one of the following alternative monetary aggregates:

- 1) M0; i.e., the monetary base, alternatively called base money, central bank money, or high-powered money. The monetary base is defined as fully liquid claims on the central bank held by the private sector, that is, currency (coins and notes) in circulation plus demand deposits held by the commercial banks in the central bank.

This monetary aggregate is under the direct control of the central bank and is changed by open-market operations, that is, by the central bank trading bonds, usually short-term government bonds, with the private sector. However, the monetary base is an imperfect measure of the

liquidity in the private sector. We will discuss more about high-powered money in Unit 6.

- 2) M1: it is defined as currency in circulation plus demand deposits held by the non-bank general public in commercial banks. M1 does not include currency held by commercial banks and demand deposits held by commercial banks in the central bank. Yet M1 includes the major part of M0 and is generally considerably larger than M0.

Broader categories of money include:

- 3) M2: M1 plus savings deposits with unrestricted access i.e., demand deposits. Although these claims may not be instantly liquid, they are close.
- 4) M3: M1 plus time deposits. M3 captures even broader liquidity in the economy.
- 5) M4: This measure includes M3 plus other forms of broad money, which may include some longer-term deposits and other less liquid financial assets like post office deposits.

4.4 MONEY AND THE PAYMENTS SYSTEM

The journey of money through history is intriguing, stretching back thousands of years. It began with simple barter systems, where goods and services were exchanged directly without a standardized medium. However, the limitations of barter, such as the difficulty in finding matching needs, led to the creation of more efficient methods of exchange.

The first significant innovation came with the introduction of metal coins. Early coins were made from bronze and copper, and later, more valuable metals like silver and gold were used. These coins represented a breakthrough because they allowed people to make payments by count rather than by weight, which simplified transactions and enabled broader commerce.

As societies and economies grew more complex, the use of coins evolved further. Alongside the physical exchange of coins, people began to develop financial instruments like claims on goods and bills of exchange. These instruments facilitated trade by providing a promise of payment or transfer of goods at a later date, thereby adding flexibility and trust to economic transactions.

The next monumental step in the history of money was the creation of paper money. Paper currency offered the advantage of being lighter and easier to transport than metal coins, which was particularly useful for large transactions and long-distance trade. This innovation marked a significant shift in how wealth was stored and transferred, and it laid the groundwork for modern banking systems.

In the present day, payment systems are undergoing rapid transformation due to technological advancements. Users now expect payments to be faster, easier, and accessible at any time, mirroring the digitalization and convenience of other aspects of life. This demand has led to the widespread adoption of near-instant person-to-person retail payment systems. These systems leverage digital technology to enable quick and seamless transactions, reflecting the continuing evolution of money and its pivotal role in society. The most recent and revolutionary phase in the evolution of money is the advent of cryptocurrency. Cryptocurrencies, such as Bitcoin and Ethereum, are digital or virtual currencies that use cryptography for security and operate on decentralized networks based on blockchain technology. They challenge traditional banking and financial systems by providing alternative ways of storing and transferring value. However, cryptocurrencies are often criticized for their extreme price volatility, regulatory uncertainty, security risks, scalability issues, energy consumption, limited acceptance, technical complexity, legal and tax implications, market manipulation, and lack of consumer protection.

Overall, the evolution of money from barter to digital payments highlights how financial innovations have consistently shaped and facilitated economic and societal development. As we move forward, the pace of change in payment systems is likely to accelerate, driven by ongoing technological advancements and the increasing expectations of consumers for convenience and speed.

Payment Mechanism-A Case of India

India also has consistently promoted innovation and development in payment and settlement systems. With the developments in the economy and the evolution of the payment system, the form and functions of money have changed over time, and it will continue to influence the future course of currency. The concept of money has experienced an evolution from commodity to metallic currency to paper currency to digital currency. The changing features of money are defining the new financial landscape of the economy. Over the past decades, numerous payment systems have emerged to benefit the common man. Since the mid-eighties, the Reserve Bank of India has introduced various technology-based solutions to enhance the banking system.

In a country like India, modern payment systems can help foster economic development and financial stability as well as support financial inclusion. The past decade has also witnessed the blossoming of quite a few payment systems that are affordable, accessible, convenient, efficient, safe, secure and available 24×7×365 days a year.

Alongside innovations in digital payments, India has implemented a separate law for Payment and Settlement Systems, to ensure the structured development of the country's payment ecosystem. The significant shift in payment preferences has resulted from the establishment of robust, round-

the-clock electronic payment systems like Real Time Gross Settlement (RTGS) and National Electronic Funds Transfer (NEFT), which facilitate seamless real-time or near real-time fund transfers. Additionally, the introduction of Immediate Payment Service (IMPS) and Unified Payments Interface (UPI) for instant payment settlements, along with mobile-based payment systems such as Bharat Bill Payment System (BBPS) and National Electronic Toll Collection (NETC) for electronic toll payments, have been transformative milestones. These advancements have attracted international recognition and ensured rapid acceptance due to the convenience they offer, providing consumers with alternatives to cash and paper-based payments. The involvement of non-bank FinTech firms as PPI issuers, Bharat Bill Payment Operating Units (BBPOUs), and third-party application providers on the UPI platform has further promoted the adoption of digital payments in India.

Furthermore, with the advent of cutting-edge technologies, the digitalization of money is the next milestone in monetary history. Advancement in technology has made it possible for the development of a new form of money viz. Central Bank Digital Currencies (CBDCs). Given the volatility and other limitations of cryptocurrency, recent innovations in technology-based payment solutions have led central banks around the globe to explore the potential benefits and risks of issuing a CBDC to maintain the continuum with the current trend in innovations. RBI also is currently engaged in working towards a phased implementation strategy, going step by step through various stages of pilots followed by the final launch, and simultaneously examining use cases for the issuance of its own CBDC (Digital Rupee (e₹)), with minimal or no disruption to the financial system. Currently, India is at the forefront of an inflexion point in the evolution of currency that will decisively change the very nature of money and its functions.

Reserve Bank broadly defines CBDC as the legal tender issued by a central bank in a digital form. It is akin to sovereign paper currency but takes a different form, exchangeable at par with the existing currency and shall be accepted as a medium of payment, legal tender and a safe store of value. CBDCs would appear as a liability on a central bank's balance sheet. The Bank for International Settlement has laid down "foundational principles" and "core features" of a CBDC, to guide exploration and support public policy objectives, as per the need of the existing mandate of Central Banks. The foundational principles emphasise that authorities would first need to be confident that issuance would not compromise monetary or financial stability and that a CBDC could coexist with and complement existing forms of money, promoting innovation and efficiency. Supported by state-of-the-art payment systems of India that are affordable, accessible, convenient, efficient, safe and secure, the Digital Rupee (e₹) system will further bolster India's digital economy, make the monetary and payment systems more efficient and contribute to furthering financial inclusion.

The dynamic and accelerated development of the payments ecosystem in India, facilitated by increased adoption of technology and innovation, has established the country as a force to reckon with in the global payments space, in terms of not only growth in digital payments but also the availability of a bouquet of safe, secure, innovative and efficient payment systems. Over 26 crore digital payment transactions are processed daily by our payment systems, of which the Unified Payments Interface (UPI) system itself processes more than two-thirds.

Check Your Progress 2

- 1) How has money evolved over the years? Highlight the role of electronic money and mobile payment systems in modern transactions.

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- 2) Discuss the evolution of payment mechanisms in India and recent innovations in the field.

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- 3) What is Central Bank digital currency? Discuss some of its potential benefits.

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4.5 MONEY, CREDIT AND THE MACROECONOMY

Understanding the intricate dynamics between money, credit, and the macroeconomy is essential for identifying how economic policies influence aggregate demand, output, and price levels.

When the demand for money rises, it reduces aggregate demand unless the monetary authority recognizes the increase in time and push up the money supply by an equal amount. Understanding money demand, and how various factors affect that demand, is, therefore, a first step in setting a target for the monetary authority. Money demand refers to the desire to hold cash or liquid assets instead of investing in less liquid assets. This demand is influenced by several factors, including income levels, interest rates, and economic uncertainty. Higher-income increases the transaction demand for money, while higher interest rates typically reduce the demand for money since